

Intelligent Task Scheduler: A Smart Productivity and Task Management System

Group Members

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Managing tasks and deadlines can be difficult, especially when people have multiple responsibilities. The Intelligent Task Scheduler is a smart task management system that helps users organize their work more effectively. It automatically prioritizes tasks based on deadlines and importance, sends reminders, and helps users stay on track. The goal of this project is to make task management easier, improve productivity, and reduce the chances of missing important deadlines.

Project Scope and Details

Project Idea

The Intelligent Task Scheduler is a web-based application designed to help users manage their tasks efficiently. The system automatically prioritizes tasks based on deadlines, importance, and estimated completion time. It helps users organize their workload, avoid missed deadlines, and improve productivity.

Project Description

Users can create tasks, set deadlines, assign priorities, and estimate the time required to complete each task. The scheduler analyzes this information and automatically arranges tasks in the most suitable order. The system also provides reminders, supports recurring tasks, and displays tasks in a calendar view.

Key Features

- User registration and login
- Create, edit, and delete tasks
- Task prioritization based on deadlines and importance
- Smart scheduling algorithm
- Reminder notifications
- Recurring tasks
- Calendar integration
- Task completion tracking
- Productivity dashboard

Why It Is Intelligent

The system automatically makes scheduling decisions instead of requiring users to organize tasks manually.

- Prioritizes urgent tasks
- Reorganizes schedules when tasks are missed
- Suggests the next task to complete

- Uses task deadlines and estimated effort to optimize schedules

Problem the Project Solves

The project addresses several common problems:

- Difficulty managing multiple tasks and deadlines.
- Forgetting important assignments, meetings, or events.
- Spending time manually organizing and prioritizing tasks.
- Poor time management leading to missed deadlines.
- Lack of a centralized system for tracking tasks and schedules.
- Challenges in balancing urgent and important work.

Week 4 – Planning and Design

- Research intelligent scheduling systems
- Finalize project requirements
- Create system architecture diagram
- Design database schema
- Create UI wireframes

Deliverable: Project proposal, architecture diagram, database design, UI mockups

Week 5 – Development

- Develop frontend interface
- Create backend APIs

- Implement database
- Develop task management features
- Implement scheduling algorithm
- Build reminder functionality

Deliverable: Functional prototype with task creation and scheduling

Week 6 – Integration, Testing, and Presentation

- Integrate frontend and backend
- Test all features
- Fix bugs and improve performance
- Prepare presentation slides
- Demonstrate the application

Deliverable: Final working application, report

Week 7–8 Deployment and Future Enhancement

Activities:

- User feedback collection
- Final documentation review
- Performance evaluation
- Security review
- Future enhancement planning
- Presentation

Team member	Roles assigned	Contribution
Amrit	Project management & system designer	System architecture, documentation, project coordination
Rishab	Backend developer	APIs, database design, authentication, task management
Nikhil	Frontend developer	User interface, dashboard, calendar view
Sampanna	Scheduling algorithm developer	Priority calculation, scheduling logic, recurring task management
Nischay	Testing & integration	Testing, debugging, integration, presentation, final report

Learning Expectations

Through this project, we hope to learn:

- Software development lifecycle
- Frontend and backend development
- Database design and management
- API development and integration
- Task scheduling algorithms
- Software testing and debugging
- Team collaboration and project management
- Building real-world productivity applications

Overall, the project:

- Stores and manages tasks
- Prioritizes important and urgent tasks
- Reminds users about upcoming deadlines
- Handles recurring tasks (daily, weekly, etc.)
- Displays tasks in an organized schedule or calendar
- Helps users improve productivity and avoid missing deadlines

Example

- Assignment due tomorrow
- Project due next week
- Club meeting on Friday

The system will automatically place the assignment at the top, remind the student about it, and organize the remaining tasks according to their deadlines and importance.

Tools and Libraries to Be Used**Frontend (UI/UX Development)**

- **HTML5** – Structure of the web application
- **CSS3** – Styling and layout design
- **JavaScript** – Interactive features and dynamic content
- **Bootstrap** (optional) – Responsive and modern UI components

Backend Development

- **Python** – Main programming language
- **Flask** – Lightweight web framework for building APIs and application logic

Database

- **SQLite** – Database for storing users, tasks, schedules, and reminders

(Can be upgraded to MySQL if needed)

Scheduling and Task Management

- **Python Datetime Library** – Date and time calculations
- **Heapq/Priority Queue** – Task prioritization and scheduling algorithms

Calendar and Notifications

- **FullCalendar.js** – Interactive calendar interface
- **Flask Mail** (optional) – Email reminders and notifications

Development and Collaboration Tools

- **Visual Studio Code (VS Code)** – Code editor
- **Git & GitHub** – Version control and team collaboration
- **Postman** – API testing
- **Figma** – UI/UX wireframes and design prototypes
- **Draw.io** – System architecture and database diagrams

Testing Tools

- **Pytest** – Backend testing
- **Browser Developer Tools** – Frontend debugging and testing

Expected Outcome

A functional Intelligent Task Scheduler that helps users manage tasks efficiently through automated prioritization, scheduling, reminders, and calendar integration while demonstrating practical software development skills.